

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
## $n.burnin
## [1] 1000
##
## $n.draw
## [1] 1200
##
## $thin
## [1] 1
##
## $pp_cutoff
## [1] 0.5
##
## $prior_gamma
## [1] 0.1 0.1 0.1 0.1 0.1
##
## $K_values
## [1] "1 / Examined number of clusters: 2" "2 / Examined number of clusters: 3"
## [3] "3 / Examined number of clusters: 4" "4 / Examined number of clusters: 5"
## [5] "5 / Examined number of clusters: 6" "6 / Examined number of clusters: 7"
## [7] "7 / Examined number of clusters: 8" "8 / Examined number of clusters: 9"
## [9] "9 / Examined number of clusters: 10"
##
## $sdev
## [1] 0.05
##
## $beta.var.scale
## [1] 0.5
```

Summary Statistics

```
--- Summary for K = 1 ---
Z.ar (Location: fit[[1]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.2133 0.2433  0.2525  0.2525 0.2608  0.2925

beta.ar (Location: fit[[1]]$beta.ar):
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.09583 0.14333 0.13956 0.17667 0.91000
```

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Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar (Location: fit[[1]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.09833 0.14667 0.14255 0.18000 0.77250
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.102124 0.030833 1.000000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.077447 0.024167 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005833 0.015000 0.162548 0.118333 1.000000

--- Summary for K = 2 ---
Z.ar (Location: fit[[2]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.04167 0.05667 0.06250 0.06281 0.06833 0.08750

beta.ar (Location: fit[[2]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0600 0.1083 0.1137 0.1492 0.8700
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar (Location: fit[[2]]$gamma.ar):
  Modality 1 :

```

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.00000	0.06333	0.11333	0.11691	0.15333	0.70250
Modality 2 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.000000	0.001667	0.058333	0.415253	1.000000	1.000000
Modality 3 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	1	1	1	1	1	1
Modality 4 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.0000000	0.0008333	0.0050000	0.2490333	0.3772917	1.0000000
Modality 5 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.0000	0.1242	1.0000	0.6980	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar (Location: fit[[3]]\$Z.ar):

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.01333	0.02750	0.03333	0.03480	0.04000	0.10083

beta.ar (Location: fit[[3]]\$beta.ar):

Modality 1 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.00000	0.03833	0.08167	0.09091	0.12250	0.89417
Modality 2 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	1	1	1	1	1	1
Modality 3 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	1	1	1	1	1	1
Modality 4 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	1	1	1	1	1	1
Modality 5 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	1	1	1	1	1	1

gamma.ar (Location: fit[[3]]\$gamma.ar):

Modality 1 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.00000	0.04000	0.08625	0.09435	0.12833	0.69917
Modality 2 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.0000	0.1583	0.4067	0.5321	1.0000	1.0000
Modality 3 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	1	1	1	1	1	1
Modality 4 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.000000	0.000000	0.001667	0.271026	0.701667	1.000000
Modality 5 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.0000	0.7917	1.0000	0.8184	1.0000	1.0000

```

--- Summary for K = 4 ---
Z.ar (Location: fit[[4]]$Z.ar):
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.003333 0.010833 0.013333 0.014171 0.016667 0.070000

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beta.ar (Location: fit[[4]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.02667 0.06583 0.07668 0.10667 0.84667
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1

```

```

gamma.ar (Location: fit[[4]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.02833 0.07000 0.08083 0.11333 0.65250
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  0.5225  0.9175  0.7147  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  0.0000  0.0000  0.2764  0.7004  1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  1.0000  1.0000  0.8754  1.0000  1.0000

```

```

--- Summary for K = 5 ---
Z.ar (Location: fit[[5]]$Z.ar):
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.001667 0.008333 0.010833 0.012332 0.015000 0.059167

```

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beta.ar (Location: fit[[5]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.01833 0.05083 0.06514 0.09083 0.77083
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1

```

```

Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar (Location: fit[[5]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.02000 0.05500 0.06905 0.09833 0.64417
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.0000  0.6083  0.5336  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.0000  0.0000  0.2611  0.7137  1.0000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  1.0000  1.0000  0.8618  1.0000  1.0000

--- Summary for K = 6 ---
Z.ar (Location: fit[[6]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0008333 0.0066667 0.0091667 0.0099680 0.0125000 0.0291667

beta.ar (Location: fit[[6]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01083 0.03500 0.05183 0.07250 0.81750
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar (Location: fit[[6]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01167 0.03833 0.05533 0.07917 0.53667
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.0000  0.0000  0.3977  1.0000  1.0000
  Modality 3 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2708	1.0000	1.0000	0.9998	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1657	0.0000	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3217	1.0000	0.7072	1.0000	1.0000

--- Summary for K = 7 ---

Z.ar (Location: fit[[7]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001667	0.006667	0.009167	0.009900	0.011667	0.050833

beta.ar (Location: fit[[7]]\$beta.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008333	0.033750	0.050328	0.071667	0.773333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar (Location: fit[[7]]\$gamma.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01000	0.03833	0.05445	0.08000	0.62417

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.3308	0.4515	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.9998	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1909	0.0000	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3175	0.9400	0.6792	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar (Location: fit[[8]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.002500	0.008333	0.010833	0.011061	0.013333	0.030000

beta.ar (Location: fit[[8]]\$beta.ar):

```

Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.00500 0.02167 0.03862 0.05583 0.71500
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

```

gamma.ar (Location: fit[[8]]\$gamma.ar):

```

Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.00500 0.02500 0.04314 0.06333 0.50750
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.0000  0.0000  0.3584  0.8883  1.0000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  1.0000  1.0000  0.9935  1.0000  1.0000
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.0000  0.0000  0.0917  0.0000  1.0000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.0000  0.3467  0.4417  0.9992  1.0000

```

--- Summary for K = 9 ---

Z.ar (Location: fit[[9]]\$Z.ar):

```

  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00500 0.01417 0.01833 0.01903 0.02333 0.04083

```

beta.ar (Location: fit[[9]]\$beta.ar):

```

Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.003333 0.021667 0.040907 0.058333 0.721667
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

```

```

gamma.ar (Location: fit[[9]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.003333 0.025000 0.046046 0.068333 0.505833
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.1255  0.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000  1.0000  1.0000  0.9935  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.0515  0.0000  1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000  0.0000  0.0075  0.2957  0.6067  1.0000

```